

More Money, More Turnout? Minimum Wage Increases and Voting

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Do minimum wage increases mobilize low-income voters? We use administrative data to measure the effect of minimum wage increases on voting behavior. We merge public records of New York City municipal employee wages to voting records to observe changes in voting by people affected and unaffected by the minimum wage across multiple elections. Difference-in-differences estimates indicate that recent increases in New York's minimum wage increased voter turnout among low-income workers by several percentage points. These findings are robust to a range of specifications and merge approaches. Further, an analysis of county-level panel data from 1980 to 2016 demonstrates that minimum wage increases are associated with increases in aggregate voter turnout across many contexts. These results imply that economic policy can have democratic implications, with minimum wage increases also serving to increase turnout among low-wage workers and make the electorate more representative.

Since 2014, 27 states and the District of Columbia have increased their minimum wages, many of them in the wake of large public campaigns and ballot measures (Economic Policy Institute 2019). We often think of such minimum wage increases as the end of a long process of political mobilization. But what if they were just the beginning of the political story?

A rich literature on “policy feedbacks” describes how policies, once enacted, can produce ongoing political action. Minimum wage increases could have such an effect in several ways, through their impacts either on economic resources or on people's views of government (Pierson 1993). People may vote more because higher incomes mean that they are better able to vote: they are less likely to encounter life crises that prevent them from voting and better able to bear the logistical costs of voting (Denny 2016; Rosenstone 1982; Verba, Schlozman, and Brady 1995). Or they could vote more because seeing the government raise the minimum wage drives home the importance of political engagement, improves their sense

of efficacy, or generates policy constituents and organizations that seek to defend the new minimum wage policy (Campbell 2003; Lawless and Fox 2001).

Although minimum wage policies are widespread and politically contested, we are aware of no attempts to empirically estimate the impact these policies have on political participation. In this project, we investigate the participatory effects of increasing the minimum wage.

We begin with a focus on individual workers. Using data on New York City municipal employees' wages and public voting records, we trace these employees' voting participation across several elections, between which there was an increase in New York's minimum wage. This design allows us to use a difference-in-differences approach: we examine changes in voter turnout among workers affected by a minimum wage increase and compare those changes to the behavior of higher-wage workers over the same time period.

We find that an hourly wage increase of up to \$1.75 increased voter turnout by several percentage points between

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2012 and 2016, compared to what we would have expected absent wage changes. These effects are also visible in midterm and mayoral elections and are robust to a range of model specifications. They are bolstered by an analysis of county-level panel data from 1980 to 2016, which takes advantage of many state minimum wage changes and also finds that wage increases are associated with higher voter turnout in the next election.

Taken together, these results paint a picture of minimum wage increases as a meaningful political force driving voting participation among low-wage workers. The analysis of individual-level municipal employee data allows us to carefully pinpoint the effect of wage increases on low-wage employees' turnout, while the panel data analysis suggests that these results generalize beyond public employees and that they are big enough to be reflected in overall turnout measures.

These results are both policy relevant and theoretically interesting. We build on recent work (e.g., Akee et al. 2018; Feigenbaum, Hertel-Fernandez, and Williamson 2019) that highlights the political importance of economic policy. More significantly, we note that minimum wage increases appear to specifically increase voting among low-wage workers, who are usually underrepresented in the electorate. This means that minimum wage policies, usually thought of as a tool to reduce economic inequality, could also serve to reduce political inequality by making the electorate more economically representative of the American people.

MINIMUM WAGE INCREASES IN NEW YORK CITY

Between the 2012 and 2016 elections, the minimum wage in New York City increased several times. A law enacted in 2013 as part of the state's budget process produced staggered increases, with the minimum wage increasing to \$8 an hour at the beginning of 2014, \$8.75 in 2015, and \$9.00 in 2016. Over this period, workers who had earned the previous minimum wage of \$7.25 per hour in 2012 received a raise of \$1.75 per hour, a 24% increase in two years.¹

Among the people seeing pay increases were thousands of city employees who had been paid below the new minimum wage rates, from school classroom aides to parks and recreation workers. These municipal employees' pay is a matter of public record, allowing us to observe any changes in these workers' voting behavior in the wake of minimum wage increases. We merge several years of detailed pay records for all hourly city employees (employees' full names, job titles, active-employee status, and pay rates) to New York voting records to see whether city employees voted in a given election. Appendix

section 2 describes this merge in detail, including results from our main deterministic merge approach as well as a probabilistic approach (Enamorado, Fifield, and Imai 2019).

The resulting sample includes 175,281 hourly New York City employees, 66% of whom match to the voter file. Consistent with the recommendations of Nyhan, Skovron, and Titunik (2017), we consider anyone who did not merge to the voter file to be a nonvoter, rather than dropping them from the sample. Observed voter turnout within the sample (unconditional on registration) was 30% in 2012 and 32% in 2016. As expected, turnout was higher among higher-wage workers: 2016 turnout among workers paid under \$12 per hour was 28%, while for workers paid more than \$12 per hour it was 39%.

Difference in differences

One way to measure a minimum wage increase's effect on voting would be to compare low-wage workers' turnout rates between 2012 and 2016. However, we might worry that any observed increase in voting over that period was not actually caused by the wage increase: maybe election differences meant that everyone voted more between 2012 and 2016, regardless of whether they were affected by the minimum wage. Rather than simply making an over time comparison, we use a difference-in-differences design. We compare the 2012–16 increase in minimum wage workers' turnout to the change in higher-wage workers' turnout (theoretically unaffected by the minimum wage increase) over the same period. This approach relies on the parallel-trends assumption; we assume that the turnout of high-wage and low-wage workers would have moved in the same ways absent a minimum wage increase. The groups' baseline levels of turnout are different, but we assume they follow parallel trends.

To determine who was affected by the minimum wage increase, we focus on employees who worked for the city before the policy change: this yields a pretreatment measure of who is affected. We identify employees who were paid below \$8 per hour any time in fiscal years 2011–14 as very likely to have been affected by the minimum wage increases that began in January 2014.² And we consider all higher-paid employees at that time “unaffected.”³ By this definition, about 17% of the hourly workers we observe over this period were treated.

2. We rely on workers who were employed with the city before the policy change so that we have a clearly pretreatment measure of who is affected, but robustness checks including 2014–16 municipal employees result in very similar estimates.

3. One could also define a continuous treatment using differences in employees' baseline pay from the new minimum wage. However (as shown in app. fig. 5), wages are so heavily clustered as to make this approach similar to the binary treatment in any given year.

1. See app. sec. 3.2 for more details on these and subsequent state/city wage changes.

After using pre-2014 wages to identify the “affected” and “unaffected” groups of workers, we then follow these two groups of workers from election to election. Rather than allowing people to drop in and out of the data set as they enter or leave municipal employment, we continue to trace the voter turnout of the same set of workers from the pretreatment election (i.e., 2012) to the postincrease election period (2016).

We regress our outcome of interest (voting) on an indicator for treatment, an indicator for the year in question (2012 or 2016), and the interaction between these two indicators. Our quantity of interest is the interaction term between the treatment (under new minimum wage) and the indicator for the 2016 (post-wage-increase) election: this tells us the amount that voting increased among affected workers due to the minimum wage change.

Figure 1 presents the main estimates from regression models based on this difference-in-differences approach, beginning with presidential election years at the top of the figure.⁴ The circles in figure 1 indicate estimates using data from all hourly municipal employees, with the first point estimate showing that the minimum wage increase boosted turnout by about 2 percentage points among affected workers between 2012 and 2016. The triangles report estimates restricting the sample to relatively low-wage workers (those making up to \$15 an hour), where we might think the parallel-trends assumption would be especially plausible.⁵ Between 2012 and 2016, this approach yields a similarly sized estimate of about 2.5 percentage points’ increase in voting by low-wage workers. We note that this difference in differences stems from a change in behavior among affected workers: unaffected workers’ turnout remained essentially flat (at 36%) between 2012 and 2016, while affected workers’ turnout increased by over 2 percentage points over the same period.

The second and third rows of the figure repeat the same exercise, this time looking at voting in midterm and off-cycle mayoral elections.⁶ The middle row of figure 1 presents difference-in-differences estimates for 2010–14, and the bottom row presents estimates for 2013–17. The estimated effects range between 1 and 3 percentage points even in these lower-turnout election years.

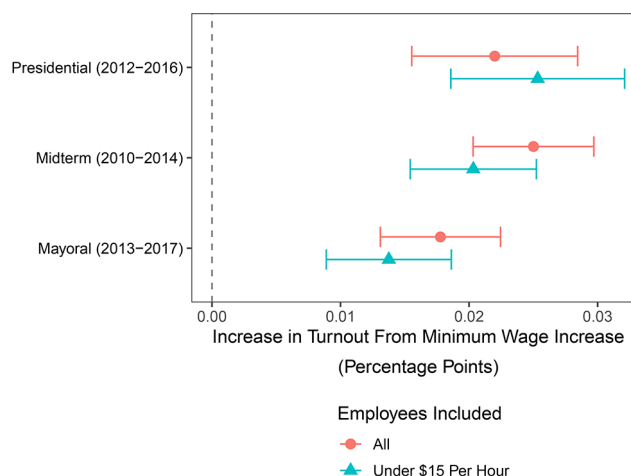


Figure 1. Individual difference-in-differences estimates

Importance and robustness of estimates

In appendix section 4, we use available estimates of how many people were affected statewide by New York’s minimum wage increases to think about the importance of these estimates. This back-of-the-envelope exercise indicates that New York state may have seen roughly 26,000 additional votes cast in 2016 because of these minimum wage increases, an increase equivalent to one-third of 1 percentage point of overall turnout statewide.

Appendix section 3 considers the assumptions of the design used here. It plots turnout trends for affected and unaffected workers over time, bolstering the parallel trends assumption invoked in our difference-in-differences analysis. It also reports the results of two different placebo tests, one based on a time period before the minimum wage increases and one based on a set of “placebo wage increases” among higher-paid workers over the same time period. Further, it reports several different specifications, including a triple-differences approach incorporating data from another state. Appendix section 3.5 (and fig. 4) considers different classifications of who is affected by the minimum wage increases and the possibility of relative status loss, while appendix section 3.6 examines the decision to treat the wage increases as a binary treatment. We also consider how merge issues could affect our estimates, and we report results from a different (probabilistic) merge approach. All these results are consistent with the findings reported here: minimum wage increases spurred higher turnout rates among low-wage municipal workers. Appendix section 3.2 discusses characteristics of the sample in more detail, paying particular attention to unemployment effects and these estimates’ generalizability to contexts beyond municipal employment in New York.

4. The regression table underlying fig. 1 appears in app. sec. 1.

5. This specification drops all workers paid more than \$15 per hour, comparing two sets of low-wage workers: one received a raise when the minimum wage went up, and one already earned slightly more than the new minimum wage. But we expect that all these workers are subject to any forces that could affect low-wage workers’ voter participation over this period.

6. We urge caution in directly comparing the size of estimates across elections, because of differences in the size and timing of wage increases and in baseline turnout. There could also be differences across time in how accurately we identify workers affected by wage increases. See app. sec. 3.2 for more on the differences between the mayoral time period and others.

Panel data

In addition to these robustness tests, appendix section 5 reports a different empirical approach, this one relying on a broader set of minimum wage increases and examining aggregate turnout rather than focusing on low-wage workers. We construct county-level data for presidential years from 1980 to 2016, combining data on federal and state minimum wage changes to produce a panel of the relevant minimum wage and presidential voter turnout rates for each county over this period. Then we run a variety of time series analyses to see whether increases in the minimum wage in a given place predict higher voter turnout in the next election.

Our main panel specification indicates that an 8% increase in the minimum wage over a given election cycle (the median increase over this period) is associated with a one-third of 1 percentage point increase in aggregate voter turnout. These findings are robust to a range of alternative specifications, including a border-county design that relies on comparing counties that fall on either side of state borders (Dube, Lester, and Reich 2010).

Unlike our main analysis, these estimates do not rely on potentially unrepresentative public employees, nor do they focus on turnout among low-wage workers. Combined with the well-identified evidence from New York shown above, they suggest that minimum wage increases have net participatory impacts large enough to be detected in overall turnout rates.

CONCLUSION

We have measured the effect of minimum wage increases on voter turnout in two different ways, and both suggest that increasing the minimum wage can also boost turnout among low-wage workers. This effect is visible at the individual level when using administrative data on wages and voting, and it is also apparent in aggregate turnout rates when using county panel data. These findings help to broaden our understanding of where policies can be expected to affect political participation. Previous work has largely focused on tangible government benefits programs like Social Security or the GI Bill; we find evidence that even nonbenefits policies can spur participation when they change people's material circumstances.⁷

These findings are consistent with a policy feedback account. But can we distinguish between “resource” or material effects of the policy and “interpretive” effects? One possible approach compares the midterm and presidential estimates presented in figure 1. That we see relatively large effects in 2014, following the partial implementation of the minimum

wage increases (only \$.75 per hour by the 2014 election) suggests that the size of the increase may be less important than its presence. Longer-term data on how quickly these effects decay—whether people keep voting more as they are still getting paid more or whether they drop off as the increases become less salient—could help fully distinguish between the resource and interpretive stories. Such data are not available in New York because of continued wage increases in recent years, but we hope that future research can disentangle these mechanisms.

Whatever the mechanism, these findings have clear political implications. Low-income Americans generally vote less than their higher-paid compatriots (Leighley and Nagler 2013), so increasing their turnout has the effect of making the electorate more representative. Given that low-wage workers also have different political attitudes than higher-paid ones, such a shift could matter for representation and future policy outcomes (Avery 2015; Franko, Kelly, and Witko 2016). Minimum wage laws are generally thought of as a tool to combat economic inequality, but they appear to also reduce political inequality along economic lines.

Ours is an era of widely divergent minimum wage laws, with some states and municipalities seeking to implement “living wage” rates while the federal minimum wage remains stagnant. We anticipate that upcoming state and local minimum wage changes will only increase these differences. This pattern could mean that electorates in some places will become more economically representative, while other places see large and growing turnout inequality along economic lines. It remains to be seen whether the patterns of voter participation we have documented here will translate into observable differences in future economic policy.

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